Stacking Galaxies Reveals a Universal Baryon Fountain

Averaging many simulated galaxies exposes smooth, regular gas flows from disks out to the cosmic web.

arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations

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Background

Galaxies grow by cycling gas inward from their surroundings and expelling it through feedback. These fountain-like flows are faint and messy in any single galaxy.

> **Individual ecosystems are too diverse and dynamic for the underlying baryon-cycle pattern to be seen directly.**

What this paper is selling

The authors stack gas velocity and property fields across ensembles of simulated galaxies, averaging away idiosyncrasies. The result is a strikingly regular fountain pattern that matches observational inferences and can be captured by a simple parameterized model.

Key results to remember

- Stacking suppresses scatter and reveals coherent inflow-outflow cones extending from disks to the intergalactic medium.
- Outflow opening angles and fountain geometry vary smoothly with stellar mass, black hole mass, and redshift.
- A compact parameterized fountain model reproduces the stacked velocity field with small residuals.

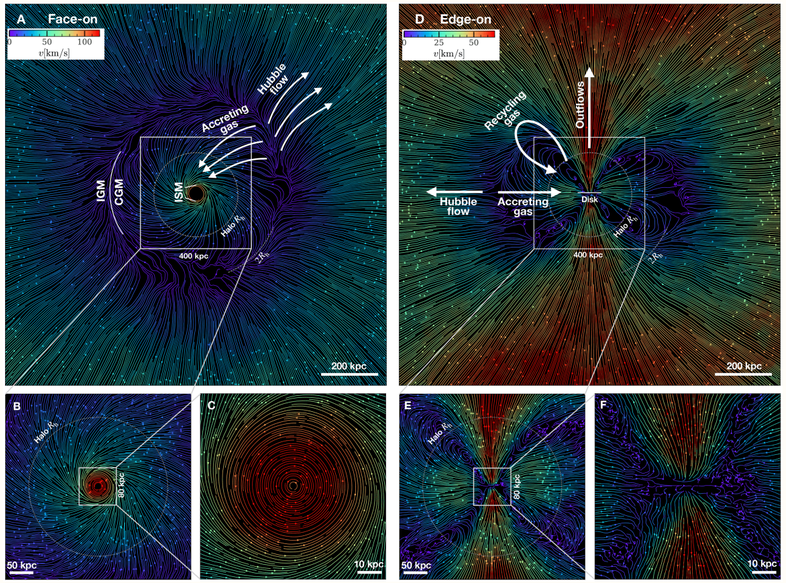


Figure 1 | Stacked streamlines show a smooth, coherent velocity field around an ensemble of Milky-Way-mass galaxies, with dashed circles marking the median virial radius.

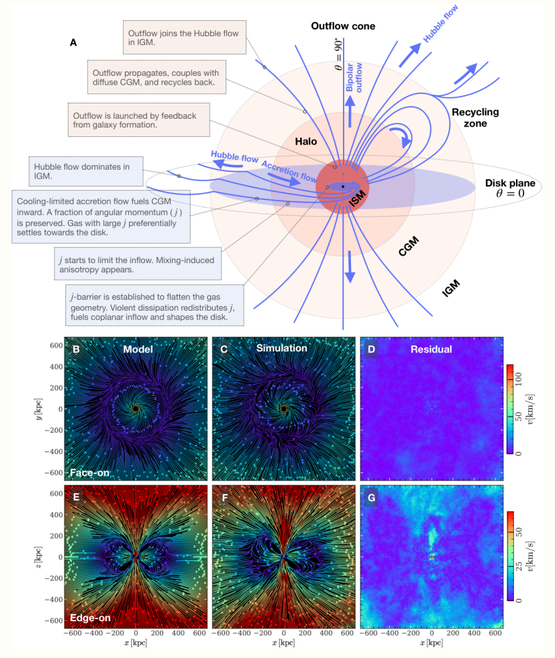


Figure 4 | A simple parameterized fountain model reproduces the stacked velocity field, leaving only small residuals and offering a compact description of the baryon cycle.